Auto-Schedule: Learning Occupant Comfort Preferences

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1. Summary

Auto-Schedule predicts the setpoint an occupant will want at a given time and applies it automatically, without a schedule programmed by the user.
Predictions come from a supervised learning model trained on manual setpoint adjustments.

2. Model

The model is a gradient-boosted decision tree (GBDT) regressor with 64 trees of depth 4.
Input features: hour of day, day of week, occupancy state, indoor and outdoor temperature.
The training target is the setpoint the occupant selected manually.

3. Learning over time

Each manual setpoint change is logged as a labelled training example.
The model is retrained nightly at 02:00 local time on a rolling 21-day window of examples.
When an occupant overrides a predicted setpoint, the override becomes a new training label.
Median prediction error falls from 1.8 °C in week one to 0.6 °C by week four
(internal field data, 12,400 homes).

4. On-device execution

Training and inference run on the thermostat's Cortex-M4F processor.
No occupant data leaves the device for model training.
Model parameters are stored in on-device flash and versioned with each retraining.

5. Relationship to occupancy sensing

Occupancy state from the PIR sensor is both a model feature and a gating signal: no predicted setpoint is applied while the home is unoccupied.

6. Scope and limitations

This whitepaper describes the algorithm as designed for firmware 5.x. Field behaviour may vary with firmware version and installation.